

E-COMMERCE ADOPTION AND ITS IMPACT ON SUSTAINABILITY IN TRADITIONAL GROCERY AND TEXTILE RETAIL IN ERNAKULAM, KERALA

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ABSTRACT

The rapid diffusion of e-commerce in India has reshaped consumer buying patterns and competitive dynamics, with particular implications for traditional grocery and textile retailers in urban districts such as Ernakulam. Recent growth in online grocery and fashion platforms, quick-commerce models, and omnichannel strategies has intensified pressure on small stores while simultaneously opening sustainability-related opportunities and risks across economic, social, and environmental dimensions[1][2][3]. This study investigates the level of e-commerce adoption among traditional grocery and textile retailers in Ernakulam and examines its perceived impact on business sustainability, including sales stability, employment, customer relationships, and adoption of sustainable practices. A structured questionnaire survey was administered to a purposive sample of 300 retailers (150 grocery, 150 textile) across major urban and peri-urban locations in Ernakulam district during November 2025 to January 2026[1][2][3]. Descriptive statistics, exploratory factor analysis, and regression models were used to analyze the relationship between e-commerce adoption intensity (independent variable) and sustainability outcomes (dependent variable). The study finds heterogeneous adoption patterns, with digitally active retailers reporting better market resilience but also facing margin pressure and logistical challenges. Economic sustainability improved significantly with higher e-commerce adoption, while environmental sustainability practices showed moderate positive association. The findings offer implications for local retailers, policymakers, and platforms seeking to align digitalization initiatives with Sustainable Development Goals (SDGs 8, 9, 11 and 12), and identify avenues for capacity-building and inclusive platform design.

Keywords: E-commerce adoption; Traditional retail; Sustainability; Grocery and textile sector; Ernakulam district

1. INTRODUCTION

Sustainability simply means living and operating in a way that can continue long-term — without depleting resources, harming ecosystems, or burdening future generations. The idea gained global recognition through the Brundtland Report (1987), which defined it as *Meeting the needs of the present without compromising the ability of future generations to meet their own needs* [52]. Over time, thinkers and policymakers realized that sustainability cannot be achieved through environment alone. A clean planet means little if people are poor, hungry, or oppressed [54]. Similarly, economic growth without social equity or environmental care is ultimately self-defeating. This led to the development of the Three Pillars Model, popularized by John Elkington (1997), often visualized as three overlapping circles - Environmental sustainability, Economic sustainability and Social sustainability [53].

Environmental sustainability means meeting present needs without compromising the ability of future generations to meet their own needs — a principle rooted in the 1987 Brundtland

Report. At its core, it's about maintaining a balance between human activity and the natural systems that support all life on Earth. Environmental Sustainability is defined as waste reduction, resource efficiency, eco-friendly packaging, consolidated delivery, and responsible inventory practices[48]. **Economic sustainability** is the capacity to maintain steady long-term growth by using resources efficiently, ensuring financial stability, promoting fair wealth distribution, encouraging innovation, and safeguarding future generations from economic or environmental burdens[49]. **Social sustainability** focuses on creating a just and inclusive society by ensuring equal access to essential services, strengthening communities, protecting human rights, preserving cultural diversity, and improving overall well-being through education and healthcare[50][51]. These three are interconnected — true sustainability requires all three working together. This is often called the "Triple Bottom Line" — People, Planet, and Profit.

India's retail sector is undergoing a structural transition as e-commerce, mobile internet, and digital payments become embedded in everyday consumption. The Indian e-commerce market has grown at an average annual rate of 18% between 2020 and 2025, while traditional retail has grown at approximately 6%, indicating a clear shift in consumer spending patterns toward digital platforms[4][5]. Online grocery and fashion segments have expanded rapidly, driven by convenience, discounts, and the emergence of quick-commerce and omnichannel models that promise instant or same-day fulfilment[6][7][8]. Traditional grocery and textile retailers—often small, family-run kirana stores and local textile outlets—face intensified competition, shifting customer expectations, and new forms of platform dependence[9][10]. The retail e-commerce market is projected to reach \$350 billion by 2030, with a compound annual growth rate (CAGR) of 23%[11].

At the same time, India's policy discourse emphasizes sustainable and inclusive growth, promoting digitalization while advancing the Sustainable Development Goals, especially those related to decent work (SDG 8), resilient cities (SDG 11), responsible consumption (SDG 12), and sustainable industry (SDG 9)[12][13]. The sustainability imperative in retail encompasses waste reduction, resource conservation, fair labor practices, and responsible sourcing[14][15].

Ernakulam district, anchored by the urban hub of Kochi, is one of Kerala's most dynamic commercial regions, with dense networks of grocery and textile stores serving diverse income groups and tourist flows. Kerala's advantageous environment of 94% literacy and 85% mobile penetration levels supported the digital transition, enabling even traditional sectors to use digital platforms[16]. The rise of e-commerce platforms and local delivery services in this district makes it an apt setting to examine how digital adoption interacts with the economic, social, and environmental sustainability of traditional retailers.

This study moves from this general national and state-level transformation to specifically investigate e-commerce penetration and its sustainability implications for grocery and textile retailers in Ernakulam district, contributing empirical evidence to an underexplored area of sustainable retail transformation in India.

2. REVIEW OF LITERATURE

2.1 E-commerce Growth and Traditional Retail Transformation

Recent studies show that online grocery adoption in India has accelerated post-pandemic, with consumers valuing convenience, safety, and digital payment options[17][18]. Karthikeyan and Vimalnath (2022) examined how the rapid growth of e-commerce is changing the landscape of

India's retail industry, highlighting that more consumers now prefer online shopping because of ease, time-saving features and lower prices, forcing traditional retailers to adapt[19].

Industry reports indicate that e-retail growth is increasingly driven by high-frequency categories such as groceries and lifestyle products, with strong expansion into smaller cities[20]. The Bain & Company (2025) report documented that online grocery shopping has witnessed strong growth in recent years, with e-commerce players expected to drive sustainability at every stage from farm to customer's doorstep[21].

Research on quick-commerce models highlights their disruptive impact on the retail ecosystem, including shifts in last-mile logistics and consumer expectations for delivery speed[22][23].

2.2 Impact on Traditional Retailers

Studies on Indian grocery retail reveal that organized and online formats pressure margins of small kirana stores but also push them toward service differentiation and digital tie-ups[24][25]. McKinsey's (2021) analysis found that traditional retailers face both challenges and opportunities in adapting to the digital shift[26].

Brick-and-mortar stores continue to thrive and grow in the Indian context despite e-commerce competition[27]. Factors influencing retailers' adoption of e-commerce include speed and reliability of delivery services, competitive pressure from online retailers, and presence of user-friendly applications[28].

2.3 Sustainability Dimensions in Retail

Work on fashion and lifestyle e-commerce suggests that sustainability considerations are beginning to influence consumer choices, especially among younger, urban customers[29][30]. E-grocery platforms are driving sustainability at scale in India, with e-commerce players sharpening their focus on both front-end and supply-chain operations[31].

Studies on sustainability practices in organized retail in India reveal that retailers implementing green store designs achieve significant operational savings, with substantial returns on investment in sustainable infrastructure[32][33]. Key sustainability areas include energy efficiency, waste management, sustainable sourcing, water conservation, recycling programs, and reducing carbon footprints[34].

2.4 Digital Transformation in Kerala's Retail Sector

Kerala's retail sector experienced significant digital transformation post-demonetization and COVID-19, with UPI transactions surging by 300% during 2020-2022[16]. The crisis prompted traditional retailers to design hyperlocal delivery networks and leverage inventory management tools, while smaller businesses adopted WhatsApp commerce and social media platforms[35][36].

Research on digitalisation in Kerala's organized retail sector identified eight dimensions of digital benefits categorized into economic, technological, and social benefits, with significant improvements in operational performance, market expansion, customer support, and employment opportunities[37].

2.5 Textile Sector and E-commerce Integration

The Indian textile and apparel market has grown from \$106 billion in 2019-20 to \$147 billion in 2024-25, registering a CAGR of 7%[41]. Digitalization and AI are progressively being

integrated into the textile sector in process control, demand forecasting, quality control, and inventory management[41].

India's digital fashion landscape has been transformed by tight planning and manufacturing flywheel based on demand sensing, rapid and flexible manufacturing, performance marketing, and just-in-time production capabilities[42]. Fashion-focused brands maintain ready stock and convert them into hundreds of options quickly based on demand trends[42].

2.6 E-commerce and Income Sustainability

Research on e-commerce and income sustainability in traditional sectors finds that digital channels can expand market access but require institutional support and capability building[41]. Trust, service quality, and perceived risk are central determinants of online grocery shopping adoption[42].

Global and Indian studies report that omnichannel strategies can help traditional retailers remain competitive while leveraging their neighbourhood proximity[43][44]. Empirical work on SME digitalization notes that lack of digital skills, infrastructure, and finance constrain e-commerce adoption, especially among micro-enterprises[45][46].

2.7 Critical Evaluation and Research Gap

Most existing studies concentrate on consumer adoption of online grocery and fashion and platform strategies, with limited micro-level evidence on how traditional grocery and textile retailers in specific districts adjust their business models and sustainability practices. Research focusing on Indian textiles often addresses export competitiveness and production sustainability but rarely examines how small urban textile retailers use or resist e-commerce channels.

Few studies systematically link degree of e-commerce adoption among small retailers to multidimensional sustainability outcomes (economic resilience, employment quality, local community ties, and environmental practices) within a single analytical framework aligned with SDGs. There is a specific gap in district-level evidence from Kerala—especially Ernakulam—on how grocery and textile retailers perceive and operationalize e-commerce, and which factors enable or hinder sustainable digital integration.

This study addresses these gaps by providing empirical evidence on e-commerce adoption patterns and their sustainability implications for traditional grocery and textile retailers in Ernakulam district.

3. STATEMENT OF THE PROBLEM

E-commerce has shifted competition in the grocery and textile markets of urban India, yet traditional retailers continue to play a critical role in local food security, employment, and cultural identity[24][26][38]. In Ernakulam, online grocery platforms, fashion marketplaces, and quick-commerce services now coexist with dense neighbourhood kirana stores and textile shops, but the extent to which these businesses are adopting digital channels—and how this affects their economic, social, and environmental sustainability—remains underexplored[16][47].

Without clear evidence, policymakers risk designing digital initiatives that inadvertently weaken local retail ecosystems, while retailers may either overinvest in inappropriate technologies or lag behind in needed capabilities. The rapid growth of e-commerce in Kerala—

with 65% of Kerala shoppers preferring digital payment types post-lockdowns—underscores the urgency of understanding how traditional retailers can sustainably integrate into the digital economy[16].

India's commitment to achieving Sustainable Development Goals by 2030, particularly SDG 8, 9, 11 and 12, necessitates empirical investigation of how digitalization in traditional retail sectors contributes to or hinders these objectives[12][13].

Accordingly, this study addresses the following research questions:

1. To what extent have traditional grocery and textile retailers in Ernakulam adopted e-commerce channels (own apps/websites, third-party platforms, social commerce)?
2. How does the level of e-commerce adoption influence their perceived business sustainability in terms of sales stability, employment, customer relationships, and environmental practices?
3. What barriers and enablers affect e-commerce adoption among traditional retailers in Ernakulam district?

These unanswered questions justify a focused empirical investigation that links e-commerce penetration with multidimensional sustainability outcomes at the district level.

4. OBJECTIVES AND HYPOTHESES

4.1 Research Objectives

1. To assess the level and patterns of e-commerce adoption among traditional grocery and textile retailers in Ernakulam district.
2. To examine the impact of e-commerce adoption on the economic, social, and environmental sustainability of these retailers.

4.2 Research Hypotheses

- **H1:** Higher levels of e-commerce adoption among traditional grocery and textile retailers in Ernakulam are positively associated with perceived economic sustainability (sales growth, customer retention, market expansion).
- **H2:** E-commerce adoption has a significant positive association with adoption of sustainability-oriented practices (reduced wastage, eco-friendly packaging, responsible inventory management).
- **H3:** There is a significant difference in e-commerce adoption levels between grocery and textile retailers in Ernakulam district.

H1 and H2 were tested using regression analysis, while H3 was tested using independent samples t-test.

4.3 Variables

- **Independent Variable (Classification of e-commerce adoption):** Level of e-commerce adoption (measured as an index based on number of digital channels used, percentage of online sales to total revenue, frequency of online transactions, and digital payment integration). It was grouped as 'low/medium/high adoption'.

- **Dependent Variables:**

- Economic sustainability (sales stability, profitability perception, customer base diversity, market reach expansion)
- Social sustainability (employment stability, skill upgrading opportunities, community relations, customer relationship quality)
- Environmental sustainability (waste reduction practices, eco-friendly packaging adoption, delivery consolidation, sustainable inventory management)

5. RESEARCH METHODOLOGY

5.1 Study Area and Rationale

The study was conducted in Ernakulam district, Kerala, covering urban and peri-urban locations including Kochi city, Tripunithura, Aluva, Kakkanad, and nearby commercial clusters. Ernakulam was selected as the study area for the following reasons:

1. **High digital penetration:** Kerala's 94% literacy rate and 85% mobile penetration provide an ideal environment for studying digital retail transformation[18].
2. **Commercial significance:** Ernakulam is Kerala's commercial capital with dense retail networks serving diverse income groups and significant tourist flows.
3. **E-commerce presence:** Visible presence of multiple e-commerce platforms, quick-commerce players, and digital payment systems makes it representative of urban digital transformation in India[49].
4. **Traditional retail density:** Strong presence of traditional kirana stores and textile outlets provides adequate sampling frame for comparative analysis.
5. **Policy relevance:** Government initiatives such as Digital Kerala Mission and support programs for SME digitalization make findings directly relevant for policy implementation[18].

5.2 Research Design and Sampling

Research Design: Descriptive and analytical, cross-sectional survey design.

Sampling Frame: Traditional brick-and-mortar grocery and textile retailers operating for at least three years in Ernakulam district, excluding purely online ventures and large-format chain stores.

Sampling Method: Purposive sampling combined with area-based quota sampling to ensure representation of both grocery and textile retailers across selected urban and peri-urban clusters.

Sample Size: 300 retailers (150 grocery retailers and 150 textile retailers). This sample size provides robust statistical power for multivariate statistical techniques including factor analysis and multiple regression, ensuring reliable and generalizable findings while remaining feasible for primary data collection.

5.3 Data Collection

Primary Data: Structured questionnaire administered through face-to-face interviews with owner-managers during November 2025 to January 2026. **34 non-adopters** were included as a comparison group to understand adoption gaps, constraints, and sustainability differences.

Secondary Data: Industry reports, government statistics, published research articles from Q1 and Q2 journals, and digital commerce platforms' publicly available data.

5.4 Research Instrument

A structured questionnaire was developed consisting of six sections:

1. Profile and business characteristics (store type, years of operation, employee count, monthly revenue range)
2. E-commerce adoption patterns (digital channels used, online sales percentage, digital payment integration)
3. Economic sustainability perceptions (7 items on 5-point Likert scale)
4. Social sustainability perceptions (6 items on 5-point Likert scale)
5. Environmental sustainability practices (6 items on 5-point Likert scale)
6. Barriers and support needs (5 items on 5-point Likert scale)

5.5 Tools of Analysis

The following statistical tools were employed:

- **Descriptive statistics:** Mean, standard deviation, frequency distribution, and cross-tabulation to profile retailers and describe e-commerce adoption levels.
- **Reliability analysis:** Cronbach's alpha to test internal consistency of multi-item scales.
- **Exploratory Factor Analysis (EFA):** To identify underlying dimensions of perceived sustainability outcomes.
- **Independent samples t-test:** To compare e-commerce adoption levels between grocery and textile retailers.
- **Multiple Linear Regression:** To test the effect of e-commerce adoption intensity on sustainability outcomes (economic, social, and environmental).
- **Data visualization:** Bar charts, pie charts, and cluster profiles for interpretation and presentation.

Statistical analysis was conducted using SPSS version 26.0 with significance level set at $p < 0.05$.

6. SCOPE, NEED, AND SIGNIFICANCE OF THE STUDY

6.1 Scope of the Study

Geographical Scope: Ernakulam district, Kerala, focusing on selected urban and peri-urban commercial locations.

Sectoral Scope: Traditional grocery retailers (kirana stores, local grocery shops) and textile retailers (fabric stores, garment shops, traditional textile outlets).

Thematic Scope: E-commerce adoption patterns, digital channel usage, business sustainability perceptions (economic, social, environmental dimensions), and alignment with Sustainable Development Goals.

Temporal Scope: Cross-sectional study conducted during November 2025 to January 2026, capturing post-pandemic digital transformation dynamics.

6.2 Need for the Study

The need for this study arises from:

1. Limited micro-level, district-specific evidence on how traditional retailers in Kerala navigate digital disruption while contributing to local economies and SDGs.
2. Policy gap in understanding which digital interventions effectively support traditional retailers without displacing them.
3. Practitioners' need for evidence-based guidance on sustainable e-commerce integration strategies.

6.3 Significance of the Study

The study holds significance for multiple stakeholders:

For Retailers: Highlights viable digital strategies and sustainability practices that support business resilience and competitive positioning in the digital era.

For Policy-makers: Informs design of interventions (digital literacy training, infrastructure support, inclusive platform models, financial assistance) that protect and upgrade local retail ecosystems while advancing SDGs.

For Platform Providers: Identifies ways to integrate traditional shops in more equitable, environmentally responsible value chains, reducing commission pressures and supporting local economies.

For Academia: Contributes empirical evidence to the emerging literature on sustainable digital transformation in traditional retail sectors in developing economies.

For Society: Contributes to understanding how digitalization can support inclusive growth, decent employment, responsible consumption, and resilient local economies aligned with India's SDG commitments.

7. LIMITATIONS OF THE STUDY

While this study provides valuable insights, the following limitations should be noted:

1. **Geographical limitation:** Findings are specific to Ernakulam district and may not be directly generalizable to rural districts, other states, or different retail categories.
2. **Purposive sampling:** Non-probability sampling limits statistical generalizability beyond the study population.
3. **Self-reported data:** Perceptions of sales, profitability, and sustainability practices may carry response bias and social desirability effects.
4. **Sectoral focus:** Limited to grocery and textile retailers; does not include other important retail segments such as pharmacies, electronics, or hardware stores.

5. **Quantitative emphasis:** While providing statistical evidence, the study does not deeply explore qualitative nuances of retailer experiences and adaptive strategies.

8. DATA ANALYSIS AND INTERPRETATION

8.1 Sample Profile

The study collected data from 300 traditional retailers in Ernakulam district, with equal representation from grocery (n=150) and textile (n=150) sectors.

Characteristic	Frequency	Percentage
Store Type		
Grocery	150	50.0
Textile	150	50.0
Years of Operation		
3-5 years	56	18.7
6-10 years	90	30.0
11-20 years	104	34.7
Above 20 years	50	16.6
Number of Employees		
1-2 employees	124	41.3
3-5 employees	116	38.7
6-10 employees	44	14.7
Above 10 employees	16	5.3
Monthly Revenue Range		
Below ₹1 lakh	70	23.3
₹1-3 lakhs	124	41.3
₹3-5 lakhs	76	25.3
Above ₹5 lakhs	30	10.0

Table 1: Sample Profile of Respondents (N=300)

8.2 E-commerce Adoption Patterns

Analysis of e-commerce adoption revealed varying levels of digital integration among traditional retailers.

Digital Channel	Grocery	Textile	Total	%
WhatsApp Commerce	104	96	200	66.7
Social Media (FB/Instagram)	76	116	192	64.0
Third-party Platforms	56	84	140	46.7
Own Website/App	24	36	60	20.0
Digital Payments Only	136	130	266	88.7
No Digital Channel	14	20	34	11.3

Table 2: Digital Channel Usage by Retailer Type (N=300)

Key Findings:

- 88.7% of retailers have adopted digital payment systems, reflecting Kerala's high UPI adoption.
- WhatsApp commerce emerged as the most popular channel (66.7%), indicating preference for low-cost, accessible platforms.
- Textile retailers show higher adoption of social media (77.3%) and third-party platforms (56.0%) compared to grocery retailers.
- Only 11.3% of retailers have no digital presence, indicating widespread but varied digital integration.
- Own website/app development remains limited (20.0%), suggesting resource constraints for advanced digital infrastructure.

E-commerce Adoption Level	Frequency	Percentage
No Adoption (0 channels)	34	11.3
Low Adoption (1 channel)	84	28.0
Medium Adoption (2 channels)	116	38.7
High Adoption (3+ channels)	66	22.0
Total	300	100.0

Table 3: E-commerce Adoption Levels (N=300)

8.3 Online Sales Contribution

Among retailers with at least one digital channel (n=266):

- Average online sales contribution: 21.3% of total monthly revenue
- Grocery retailers: 16.8% average online sales contribution
- Textile retailers: 26.4% average online sales contribution
- Range: 5% to 45% of total revenue from digital channels

Textile retailers derive significantly higher revenue share from online channels compared to grocery retailers, likely due to wider catchment areas, visual presentation advantages, and lower frequency of purchase.

8.4 Reliability Analysis

Cronbach's alpha was calculated for multi-item scales:

Scale	Items	Cronbach's Alpha
Economic Sustainability	7	0.86
Social Sustainability	6	0.82
Environmental Sustainability	6	0.79
E-commerce Adoption Index	5	0.88

Table 4: Reliability Statistics for Research Scales

All scales demonstrated high internal consistency ($\alpha > 0.79$), confirming reliability of measurement instruments.

8.5 Impact on Economic Sustainability

Multiple regression analysis was conducted with e-commerce adoption index as the independent variable and economic sustainability score as the dependent variable.

Model	R	R ²	Adjusted R ²	Std. Error
Economic Sustainability	0.68	0.46	0.45	0.52

Table 5: Model Summary: E-commerce Adoption Impact on Economic Sustainability

Variable	B	Std. Error	Beta	t	Sig.
(Constant)	2.18	0.15	-	14.53	0.000
E-commerce Adoption	0.72	0.06	0.68	16.97	0.000

Table 6: Regression Coefficients: Economic Sustainability

Interpretation:

The regression model is statistically significant ($F = 288.00$, $p < 0.001$). E-commerce adoption explains 46% of variance in economic sustainability perceptions ($R^2 = 0.46$). The positive coefficient ($\beta = 0.68$, $p < 0.001$) supports **H1**, indicating that higher e-commerce adoption is significantly associated with improved perceived economic sustainability.

Retailers with high e-commerce adoption reported:

- 58% improvement in sales stability
- 52% expansion in customer base beyond local neighborhood
- 45% better ability to compete with larger retailers
- 38% improvement in long-term business prospects

8.6 Impact on Social Sustainability

Multiple regression analysis was conducted with e-commerce adoption index as the independent variable and social sustainability score as the dependent variable. The results indicate that e-commerce adoption has a significant positive effect on social sustainability

among traditional grocery and textile retailers in Ernakulam district. The model explains a meaningful proportion of variance in social sustainability, suggesting that digital adoption contributes to employment stability, skill development, customer relationship quality, and community engagement.

Model	R	R ²	Adjusted R ²	Std. Error
Social Sustainability	0.49	0.24	0.23	0.61

Table 7: Model Summary: E-commerce Adoption Impact on Social Sustainability

Variable	B	Std. Error	Beta	t	Sig.
(Constant)	2.31	0.16	—	14.44	0.000
E-commerce Adoption	0.54	0.07	0.49	7.71	0.000

Table 8: Regression Coefficients for E-commerce Adoption and Social Sustainability

Interpretation:

The regression model is statistically significant, indicating that e-commerce adoption has a meaningful positive effect on social sustainability among traditional grocery and textile retailers. The positive coefficient supports H2, showing that higher e-commerce adoption is associated with better employment stability, stronger customer relationships, improved skill development, and greater community engagement.

Retailers with high e-commerce adoption reported:

- 46% improvement in employment stability.
- 41% increase in customer relationship quality.
- 39% better staff skill development.
- 34% stronger community interaction through digital channels.

8.8 Impact on Environmental Sustainability

Model	R	R ²	Adjusted R ²	Std. Error
Environmental Sustainability	0.42	0.18	0.17	0.68

Table 9: Model Summary: E-commerce Adoption Impact on Environmental Sustainability

Variable	B	Std. Error	Beta	t	Sig.
(Constant)	2.65	0.18	-	14.72	0.000
E-commerce Adoption	0.38	0.08	0.42	6.72	0.000

Table 10: Regression Coefficients: Environmental Sustainability

Interpretation:

The model is statistically significant ($F = 45.12$, $p < 0.001$). E-commerce adoption explains 18% of variance in environmental sustainability practices ($R^2 = 0.18$). The positive coefficient ($\beta = 0.42$, $p < 0.001$) provides partial support for **H2**, indicating moderate positive association between e-commerce adoption and environmental sustainability practices.

Among high adopters:

- 34% use eco-friendly or reduced packaging for deliveries

- 28% report reduced product wastage through better demand forecasting
- 22% attempt delivery consolidation to minimize fuel use
- 18% actively source sustainable products for online customers

The moderate relationship suggests that while e-commerce adoption creates opportunities for sustainable practices, most retailers have not yet fully leveraged these possibilities.

8.9 Comparison Between Grocery and Textile Retailers

Independent samples t-test was conducted to test **H3**.

Sector	N	Mean	Std. Dev.	t-value	Sig.
Grocery	150	2.18	0.85	-4.84	0.000
Textile	150	2.68	0.92		

Table 11: E-commerce Adoption Index by Sector

Interpretation:

There is a statistically significant difference in e-commerce adoption levels between grocery and textile retailers ($t = -4.84$, $p < 0.001$), supporting **H3**. Textile retailers demonstrate significantly higher e-commerce adoption ($M = 2.68$, $SD = 0.92$) compared to grocery retailers ($M = 2.18$, $SD = 0.85$).

This difference can be attributed to:

- Nature of products: Textiles are less perishable and easier to ship
- Visual presentation: Textiles benefit from photo-based online marketing
- Market reach: Textiles attract customers from wider geographic areas
- Purchase frequency: Lower frequency allows time for online discovery and consideration

9. FINDINGS AND CONCLUSION

9.1 Key Findings

The study yields the following major findings:

1. **Enhance digital literacy programmes.** Given that inadequate digital skills remain a primary obstacle and that greater adoption contributes positively to both economic and social sustainability, local trade associations and government bodies should deliver accessible, hands-on training covering WhatsApp commerce, Instagram-based selling, digital payment systems, and online customer service management.
2. **Prioritise affordable digital entry points.** Since the majority of small retailers already rely on WhatsApp and digital payments rather than dedicated websites, they should be guided to start with low-cost solutions such as WhatsApp Business, Google Business Profile, Instagram product catalogues, and UPI-integrated billing systems before transitioning to more sophisticated platforms.

3. **Tailor strategies according to retail sector.** As textile retailers demonstrate comparatively higher digital adoption and stronger social sustainability outcomes than their grocery counterparts, textile businesses should concentrate on visual content marketing, social media-driven sales, and broader customer outreach, whereas grocery retailers should prioritise streamlined ordering processes, repeat purchase convenience, and efficient local delivery systems.
4. **Provide operational support to grocery retailers.** Considering that grocery retail faces unique constraints related to product perishability and high-frequency delivery demands, these businesses should receive targeted assistance in areas such as inventory management, demand forecasting, consolidated delivery planning, and partnerships with local logistics providers to enhance both digital adoption and sustainability performance.
5. **Strengthen social sustainability through digital engagement.** Since regression findings confirm a positive association between e-commerce adoption and social sustainability, retailers should leverage digital platforms not merely as sales channels but also as tools for employee upskilling, customer relationship building, and the development of community-oriented service approaches.
6. **Advocate for equitable platform policies.** As excessive platform fees and over-dependence on intermediaries continue to constrain adoption among small retailers, policymakers should work towards establishing fair commission frameworks, providing structured onboarding support, ensuring transparent seller guidelines, and promoting inclusive digital marketplace models such as ONDC-style access to lower entry barriers.
7. **Apply environmental sustainability measures deliberately.** Since digitalisation yields only a modest improvement in environmental sustainability, retailers should be actively encouraged to implement targeted practices including minimised packaging, consolidated delivery routes, optimised logistics planning, and systematic waste reduction, rather than assuming that digital adoption alone will generate meaningful ecological benefits.

9.2 Conclusion

E-commerce adoption among traditional grocery and textile retailers in Ernakulam district represents an ongoing transformation rather than a complete disruption. The study demonstrates that digital integration can enhance the resilience and sustainability of traditional retailers when accompanied by appropriate capabilities, resources, and support systems.

The finding that e-commerce adoption significantly improves economic sustainability validates the potential of digitalization to strengthen rather than replace traditional retail. However, the moderate environmental sustainability outcomes indicate that digital transformation alone does not automatically lead to greener retail practices—intentional interventions are needed.

The significant sectoral differences underscore the importance of context-specific digital strategies. Textile retailers' higher adoption and revenue contribution suggest that product characteristics, customer behavior, and operational models shape digital integration trajectories.

For traditional retailers to achieve sustainable digital transformation aligned with SDGs 8, 9, 11, and 12, three enabling conditions must be met:

- **Capability building:** Digital literacy programs, technical training, and mentorship to overcome skill barriers.
- **Inclusive platforms:** Fair commission structures, transparent policies, and local retailer integration models that preserve margins and community relationships.
- **Sustainability support:** Access to eco-friendly packaging, logistics optimization tools, and incentives for green practices.

The study concludes that e-commerce and traditional retail need not be adversarial; with appropriate policies and practices, digital channels can serve as enablers of sustainable, inclusive local retail ecosystems in urban India.

10. IMPLICATIONS AND RECOMMENDATIONS

10.1 Implications for Policy-makers

1. Provide financial subsidies or low-interest loans for digital infrastructure investments (POS systems, inventory management software, website development).
2. Facilitate partnerships between traditional retailers and e-commerce platforms through regulatory incentives that encourage fair commission structures and local integration.
3. Develop sustainability certification and incentive schemes for retailers adopting eco-friendly packaging, waste reduction, and responsible sourcing practices.
4. Strengthen last-mile logistics infrastructure in peri-urban areas to reduce delivery costs and environmental impacts.

10.2 Implications for Retailers

1. Start with accessible digital channels (WhatsApp Business, Instagram, Facebook) before investing in expensive infrastructure.
2. Explore omnichannel strategies (online discovery + offline experience) that combine digital reach with physical store strengths.
3. Invest in basic digital literacy and train staff in online customer service and digital payment systems.
4. Collaborate with other local retailers to share logistics costs and negotiate better platform terms.
5. Proactively adopt sustainable practices (eco-friendly packaging, waste reduction) as differentiators in digital marketplace.

10.3 Implications for Platform Providers

1. Provide free or subsidized training, technical support, and onboarding assistance for traditional retailers.
2. Create local retailer integration features that preserve neighborhood identity and community relationships.
3. Offer sustainability tools (packaging options, delivery consolidation algorithms, waste tracking) within platform interface.

11. SCOPE FOR FURTHER RESEARCH

Future research could extend and deepen insights from this study through:

1. **Longitudinal studies:** Track retailers' digital transformation trajectories and sustainability outcomes over 3-5 years to capture evolutionary dynamics and long-term impacts.
2. **Geographic expansion:** Extend analysis to rural districts, other states, and comparative studies across regions to identify context-specific patterns and generalizable insights.
3. **Sectoral diversification:** Include other retail categories (pharmacies, electronics, hardware, restaurants) to develop comprehensive understanding of sector-specific digital transformation patterns.
4. **Consumer perspective:** Incorporate consumer surveys to understand demand-side drivers of sustainable retail and willingness to pay for eco-friendly practices.
5. **Qualitative depth:** Conduct in-depth case studies and ethnographic research to capture rich narratives of retailer adaptation strategies, challenges, and innovations.
6. **Platform governance:** Analyze platform-side policies, algorithms, and business models to identify mechanisms for more equitable and sustainable digital retail ecosystems.
7. **Impact measurement:** Develop robust frameworks for measuring actual (not just perceived) sustainability outcomes including carbon footprints, waste generation, employment quality, and community impacts.
8. **Policy evaluation:** Assess effectiveness of government interventions (Digital Kerala Mission, ONDC, SME digitalization programs) in supporting sustainable retail transformation.
9. **Comparative analysis:** Compare Kerala's experience with other Indian states and international contexts to identify transferable lessons and unique contextual factors.
10. **Technology adoption:** Investigate emerging technologies (AI-based inventory management, blockchain for supply chain transparency, IoT for waste reduction) and their applicability to traditional retail sustainability.

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